On Feature Decorrelation in Self-Supervised Learning

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(Preliminaries) Invariance Learning in Self-Supervised Learning

- **SSL approaches can be categorized into 3 groups.**
  - Let X be data, $Z(X)$ be representation, and Y be pretext label.
  - Let I denotes mutual information of two random variables.
    - **Pretext task:** Maximize $I(Z(X); Y)$
    - **Invariance**: Maximize $I(Z(X_1); Z(X_2))$ where X_1, X_2 are invariant data.
    - **Generation:** Maximize $I(Z(\bar{X}); X)$ where $\bar{X}$ is perturbed version of X.

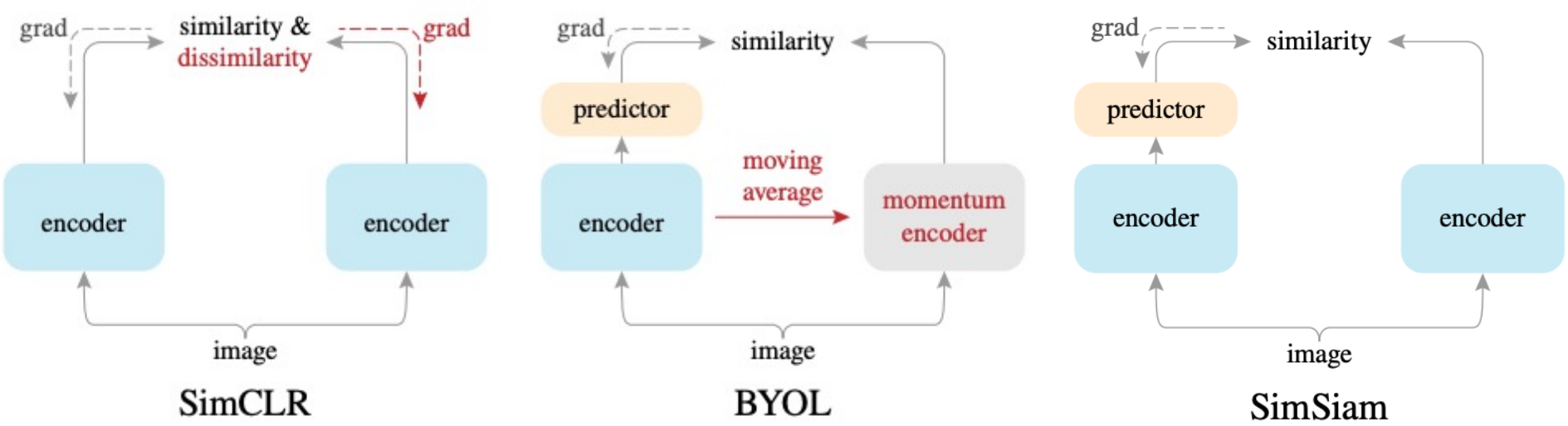
- **Core idea of invariance-based Learning**
  - Invariance: Representations of positive pairs should be similar.

Positive pair $f($ $) \approx f($ $)$

(Preliminaries) Problems of Invariance Learning

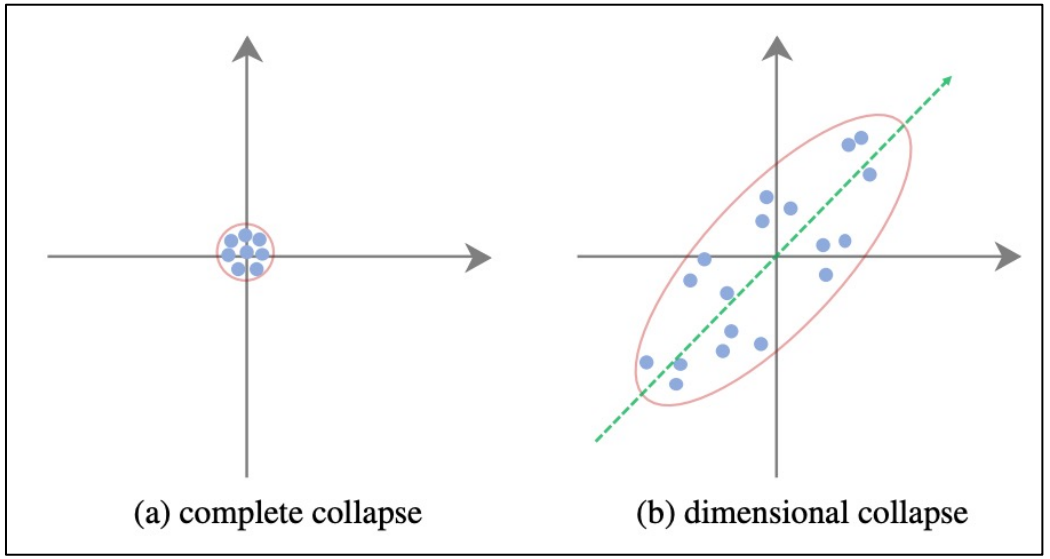
- **Can we learn representations by just simply training a positive pair to be similar?**
  - Simply minimizing $|f($ $) - f($ $)|$ leads to **mode collapse**.
  - Mode collapse: $\forall x, f(x) = c$

- **Solution**
  - **SimCLR:** use **contrastive learning** to repulse samples from different images.
  - **BYOL:** Use **predictor with batch-normalization** and **momentum encoder**.
  - **Simsiam:** Use **bottleneck predictor with batch-normalization**.



(Motivation) Overlooked Collapse Pattern: Dimensional collapse

- **Complete Collapse vs Dimensional Collapse**



- Complete collapse:
  representations collapse in a single point.
- Dimensional collapse:
  representations collapse in a few dimension.

- **Does batch-norm prevents dimensional collapse?**

Dimensional collapse remains

x_1	1	2
x_2	1.1	2.2
x_3	0.9	1.8

→ Batch normalization (affine=False) →

x_1	0	0
x_2	1.25	2.5
x_3	-1.25	-2.5

Model: Concise Framework

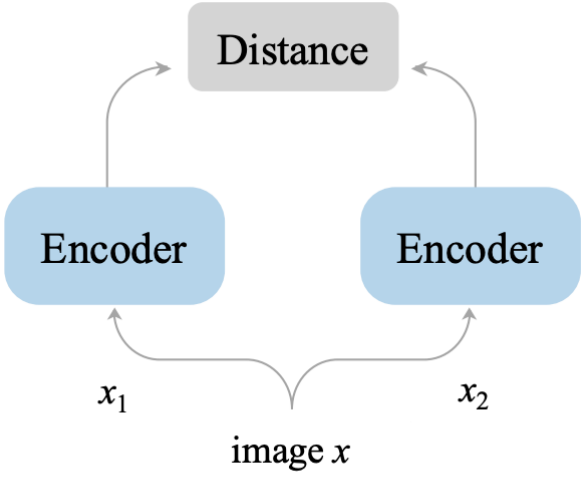
- **Main idea:** Use **Decorrelated Batch Normalization** to prevent dimensional Collapse.

- **Model architecture**

Definition 1 (Concise Framework) *In the concise framework, given the training data distribution $\mathcal{D}$ and the augmentation distribution $\mathcal{T}$, the model parameter θ is trained to maximize/minimize the objective function with the following form:*

$$\mathcal{L}(\theta) = \mathbb{E}_{\substack{x \sim \mathcal{D} \\ T_1, T_2 \sim \mathcal{T}}} \ell(f_\theta(x_1), \ f_\theta(x_2)),$$

where f_θ is the encoder that contains a backbone and a projector, $x_1 = T_1(x)$, $x_2 = T_2(x)$, and ℓ is the similarity/distance function. A sketch of this framework is presented in Figure 1d.



(d) the concise framework

ZCA Whitening

Definition 4 (ZCA Whitening [3]) *ZCA Whitening takes as its input a batch of D-dimensional vectors $X = (x_1, \cdots, x_B) \in \mathcal{R}^{D \times B}$ and its output is a batch of vectors $Y = (y_1 \cdots, y_B) \in \mathcal{R}^{D \times B}$ computed as follows:*

$$Y = Q\Lambda^{-\frac{1}{2}}Q^T\hat{X},$$

where $\hat{X}$ is X with rows normalized to zero-mean (i.e. $\hat{X}_{d,b} = X_{d,b} - \frac{1}{B}\sum_{k=1}^{B} X_{d,k} = x_{b,d} - \frac{1}{B}\sum_{k=1}^{B} x_{k,d}$), $\Lambda \in \mathcal{R}^{D \times D}$ is a diagonal matrix filled with the eigenvalues of $\Sigma = \hat{X}\hat{X}^T$ and $Q \in \mathcal{R}^{D \times D}$ is the corresponding orthonormal eigenvectors (i.e. $\Sigma = Q\Lambda Q^T$). ZCA assumes $\Sigma = \hat{X}\hat{X}^T \in \mathcal{R}^{D \times D}$ is full-rank.

1. Standardize the input to zero mean.
2. Compute Covariance matrix. $\Sigma = \hat{X}\hat{X}^T$
3. Perform SVD to Covariance matrix. $\Sigma = Q\Lambda Q^T$
4. Perform ZCA whitening to X.

- ZCA Whitening

$$\begin{aligned} YY^T =& Q\Lambda^{-\frac{1}{2}}Q^T\hat{X}\hat{X}^TQ\Lambda^{-\frac{1}{2}}Q^T \\ =& Q\Lambda^{-\frac{1}{2}}Q^TQ\Lambda Q^TQ\Lambda^{-\frac{1}{2}}Q^T \\ =& Q\Lambda^{-\frac{1}{2}}\Lambda\Lambda^{-\frac{1}{2}}Q^T \\ =& QQ^T = I. \end{aligned}$$

ZCA Whitening transforms the representations to let the covariance matrix matches the identity matrix.

Decorrelated Batch Normalization

Definition 3 (Decorrelated Batch Normalization [25]) *The Decorrelated Batch Normalization (DBN) layer with a group size G takes as its input a batch of D-dimensional vectors $X = (x_1, \cdots, x_B) \in \mathcal{R}^{D \times B}$ and its output is a batch of vectors $Y = (y_1 \cdots, y_B) \in \mathcal{R}^{D \times B}$ computed as follows:*

$$Y^{[h]} = ZCA(X^{[h]}),$$

where $X^{[h]} = \left(\left(X_{(h-1)\cdot G+1}\right)^T, \cdots, \left(X_{h \cdot G}\right)^T \right)^T \in \mathcal{R}^{G \times B}$ *and* $Y^{[h]} = \left(\left(Y_{(h-1)\cdot G+1}\right)^T, \cdots, \left(Y_{h \cdot G}\right)^T \right)^T \in \mathcal{R}^{G \times B}$. *In other words, DBN divides the D feature dimensions into groups of size G and applies ZCA whitening to each group independently.*

3.4.3 The Role of Grouping in DBN

The grouping strategy in DBN has two major benefits, one is for flexibility, and another is for efficiency.

Recall that ZCA Whitening works only with the assumption that $\Sigma = \hat{X}\hat{X}^T \in \mathcal{R}^{D \times D}$ is full-rank. Otherwise, no linear transform on the features can result in a fully standardized covariance matrix, *i.e.* an identity matrix I, since I is full-rank.

To have Σ a matrix with rank D, $\hat{X} \in \mathcal{R}^{D \times B}$ must have a rank of at least D. Besides, since each row of $\hat{X}$ is zero-mean, we have its rank to be bounded by $B - 1$, which indicates that the minimum batch size B allowed is at least $D + 1$. This greatly limits the flexibility of ZCA Whitening as one has to either restrict the dimension of the feature space or scale the batch size linearly with it. With grouping, the batch size only has to scale with the group size G.

Loss function: MSE

3.4.1 The Choice of The Objective ℓ

Here we will explain why we choose squared error $\ell_{SE}(z_1, z_2) = \|z_1 - z_2\|_2^2$ as our default setting instead of cosine similarity $\ell_{cos}(z_1, z_2) = \frac{z_1^T z_2}{\|z_1\|_2 \|z_2\|_2}$, which is more popular in self-supervised representation learning.

The main difference between maximizing cosine similarity ℓ_{cos} and minimizing squared error ℓ_{SE} is whether or not the vectors are normalized to unit L_2-norms, since $\ell_{SE}(z_1, z_2) = 2 - 2\ell_{cos}(z_1, z_2)$ when $\|z_1\|_2 = \|z_2\|_2 = 1$.

Such normalization may conflict with feature decorrelation. With squared error, for $i \in \{1, 2\}, j \in \{1, \cdots, D\}$ where $z_1, z_2 \in \mathcal{R}^D$, we have

$$\frac{\partial \ell_{SE}}{\partial z_{i,j}} = 2(z_{i,j} - z_{3-i,j}),$$

which involves only the j-th dimension itself.

However, with cosine similarity, for $i \in \{1, 2\}, j \in \{1, \cdots, D\}$ where $z_1, z_2 \in \mathcal{R}^D$, we have

$$\frac{\partial \ell_{cos}}{\partial z_{i,j}} = \frac{z_{3-i,j}}{\|z_{3-i}\|_2} \cdot \left(\frac{1}{\|z_i\|_2} - \frac{z_{i,j}^2}{\|z_i\|_2^3} \right) = \frac{z_{3-i,j} \sum_{k \neq j} z_{i,k}^2}{\|z_{3-i}\|_2 \|z_i\|_2^3},$$

which may depend heavily on other dimensions because of the normalization. This may introduce unnecessary interference in the projection space.

Validating Assumption: Dimensional collapse occurs in 2d representations.

- **Experimental Setup**
  - Dataset: CIFAR-10
  - Encoder: Resnet-18 where output dimension: 2
  - Projector: 2-layer MLP with 64 dimensions

- **Results**

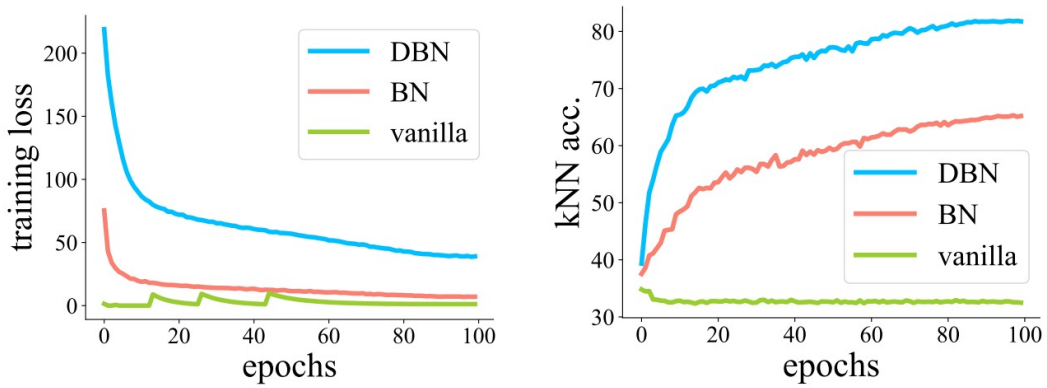


Figure 3: A comparison of learning processes with different variants of the concise framework. For both collapse patterns, collapsed variants optimize loss function easily but offer representations with degraded utility.

	acc. (%)	std.	corr.	loss
vanilla	35.44	**0.00**	0.13	0.00
BN	70.85	1.00	**0.99**	7.01
DBN	84.41	1.00	0.00	39.04

Table 1: A comparison of the eventual representations with different variants of the concise framework: **acc.** denotes the accuracy in linear evaluation; **std.** denotes the average standard deviation over 128 dimensions of the projected features; **corr.** denotes the average correlation strength (*i.e.* the average of the absolute values of non-diagonal entries of the correlation matrix) of the projected features; **loss** denotes the training loss. The group size of DBN is 128.

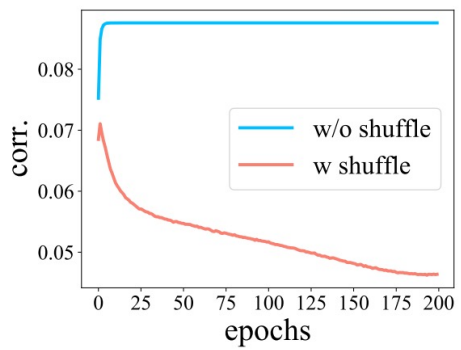
Further Decorrelation: Shuffled DBN

Definition 5 (Shuffled-DBN) *The Shuffled-DBN layer with a group size G takes as its input a batch of D-dimensional vectors $X = (x_1, \cdots, x_B) \in \mathcal{R}^{D \times B}$ and its output is a batch of vectors $Y = (y_1 \cdots, y_B) \in \mathcal{R}^{D \times B}$ computed as follows:*

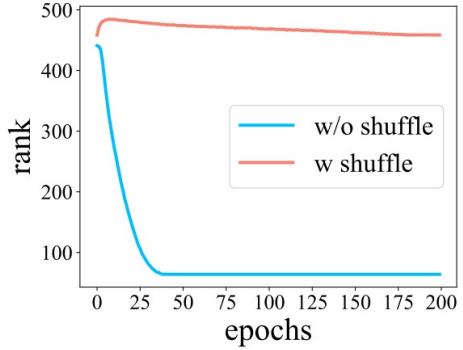
$$Y = \mathcal{P}^{-1}\left(DBN_G(\mathcal{P}(X))\right),$$

where $\mathcal{P}$ is a random D-order permutation, $\mathcal{P}(X)$ is obtained by rearranging rows of X according to $\mathcal{P}$ and $\mathcal{P}^{-1}(X)$ is obtained by rearranging rows according to the inverse permutation of $\mathcal{P}$.

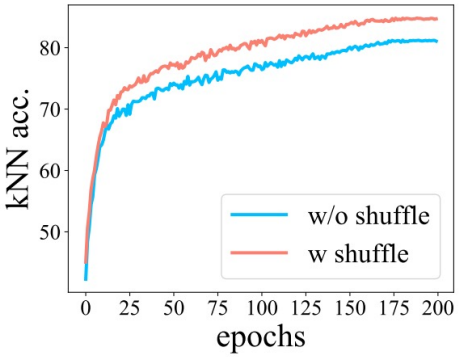
In other words, Shuffled-DBN permutes the D feature dimensions randomly before applying DBN with the same group size G and reverses the permutation for outputs.



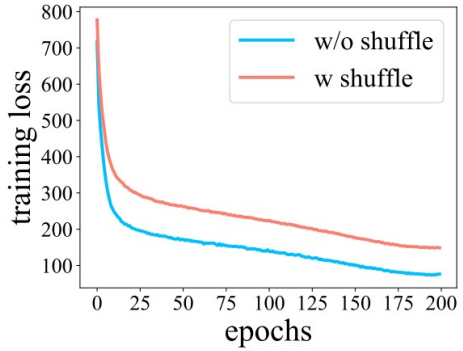
(a) **corr.** denotes the average correlation strength (*i.e.* the average of the absolute values of non-diagonal entries of the correlation matrix) of the projected features.



(b) **rank** denotes the (estimated) rank of spaces spanned by projected features of 512 samples, which is computed by checking singular values.



(c) **acc.** denotes accuracy in kNN classification.



(d) **loss** denotes the training loss.

Shuffled-DBN

: just random selection of rows for each group.

Experiments

- Linear Evaluation in Small Dataset

	CIFAR-10	CIFAR-100	STL-10	Tiny ImageNet
SimCLR [8]	86.96	55.86	85.50	42.65
BYOL [37]	86.65	59.33	85.59	42.75
SimSiam [10]	86.31	59.44	**86.55**	41.58
Barlow Twins [51]	89.02	62.84	85.43	45.33
DBN	86.32	56.49	82.36	40.37
Shuffled-DBN	**89.50**	**62.95**	86.02	**45.96**

- Transfer Learning in Small Dataset

	CIFAR-10	CIFAR-100
SimCLR	75.05	50.42
BYOL	78.63	51.44
SimSiam	78.77	50.71
Barlow Twins	79.54	**57.22**
DBN	78.60	52.95
Shuffled-DBN	**80.62**	57.17

Table 4: Top-1 accuracies(%) of DBN and Shuffled-DBN in linear evaluation on CIFAR-10 and CIFAR-100 with 200-epoch pretraining on Tiny ImageNet. For completeness and reference, we include results of some representative methods from our reproduction. For a fair comparison, we use the same projector and augmentations as we describe in Section 4.1 for all methods in the reproduction.

Experiments

- ImageNet

method	batch size	top-1
InstDisc [46]	256	58.5
LocalAgg [52]	128	58.8
MoCo [19]	256	60.6
SimCLR [8]	256	61.9
CPC v2 [35]	512	63.8
PCL v2 [33]	256	67.6
MoCo v2 [9]	256	67.5
MoCHi [29]	512	68.0
PIC [4]	512	67.6
AdCo [24]	256	68.6
Shuffled-DBN	512	65.18

Table 8: Top-1 accuracies(%) in linear evaluation on ImageNet with the ResNet-50 backbone and 200 epochs of pretraining. The table are mostly inherited from [24]. Feature decorrelation in the concise framework acheives sub-optimal performance.

On ImageNet, although Shuffled-DBN does not achieve state-of-the-art performances, it remains promising given that it achieves sub-optimal utility in a concise framework (*i.e.*, with no predictor, no momentum encoder and no other special implementation detail).

- Ablation on Group Size

group size	16	32	64	128
kNN acc.	83.41	85.93	87.05	87.59
linear acc.	85.52	87.69	88.75	88.29

Table 7: Accuracies(%) of Shuffled-DBN on CIFAR-10 with different group size. **kNN acc.** denotes accuracy in kNN classification. **linear acc.** denotes accuracy in linear evaluation. The output dimension of the projector is 512.

Conclusion

- **Personal Thoughts**
  - Preventing Dimensional Collapse with DBN can be easily adopted in SSL.
  - The idea is pretty much similar to the concurrent work Barlow Twins.
  - It is interesting that Barlow Twins and DBN performed best in transfer-learning.
    - (Higher rank implies better transferability?)

Q & A